FF No.: 654

AI2003: COMPUTER NETWORK

Course Prerequisites: Fundamentals of Computer, C/C++ programming.

Course Objectives:
1. Understand the importance of Computer Network and its usage.
2. Study error control and flow control techniques.
3. Solve real-world problems in the context of today's internet (TCP/IP and UDP/IP).
4. Distinguish and relate various physical Medias, interfacing standards and adapters.
5. Implement mathematically and logically the working of computer protocols in abstract.

Credits:4

Teaching Scheme Theory: 2 Hours/Week
Tut: 1 Hours/Week
Lab: 2 Hours/Week

Course Relevance:
A system of interconnected computers and computerized peripherals such as printers is called computer network. This interconnection among computers facilitates information sharing among them by using data communication. The main objective of computer network is to enable seamless exchange of data between any two points in the world. This course will explore common network services and protocols such as email, web services etc Networking is an ever growing domain in which there is a constant need of support. Networks are becoming progressively more and more convoluted as the technology is advancing and flourishing.

Section 1
Introduction:Introduction to computer network, LAN, MAN, WAN, PAN, Ad-hoc Networks,Network Architectures- Client-Server, Peer To Peer, Network Topologies- Bus, ring, tree, star, mesh,hybrid. Communication Models- OSI Model, TCP/IP Model, Design issues for layers. Physical Layer:Transmission media- Guided media, unguided media. Transmission Modes- Simplex, Half-Duplex and Full-Duplex. Network Devices- Hub, Repeater, Bridge, Switch, Router, Gateways and Brouter. Spread spectrum signal, FHSS, DSSS. Data Link Layer:Logical Link Layer- Services to Network Layer, Framing, Error Control and FlowControl. Framing in LLC- framing challenges, types of framing. Error Control in LLC- error detection, error correction, Parity Bits, Hamming Codes (11/12-bits) and CRC. Flow Control Protocols- Unrestricted Simplex, Stop and Wait, Sliding Window Protocol. WAN Connectivity- PPPand HDLC. Medium Access Control: Channel Allocation-Static and Dynamic, Multiple Access Protocols- Pureand Slotted ALOHA, CSMA, WDMA, IEEE 802.3 Standards and Frame Formats, CSMA/CD.
Section 2

Network Layer: Switching techniques, IP Protocol, IPv4 and IPv6 addressing schemes, Subnetting, NAT, CIDR, ICMP, Routing Protocols- Distance Vector, Link State, Path Vector, Routing inInternet- RIP,OSPF, BGP, Congestion control and QoS,
Transport Layer: Services, Berkley Sockets, Addressing, Connection establishment, Connection release, Flow control and buffering, Multiplexing, TCP, TCP Timer management, Quality of Service (QoS), Differentiated services, TCP and UDP for Wireless.
Application Layer: Domain Name System (DNS), Hyper Text Transfer Protocol (HTTP), Email:SMTP, MIME, POP3, Webmail, FTP, TELNET, Dynamic Host Control Protocol (DHCP), Simple
Network Management Protocol (SNMP).

List of Tutorials: (Any Three)

1. Identification of various networks components
2. Establishing LAN
3. Installation of network device drivers
4. Use/installation of proxy server
5. Configuration of network devices in CISCO packet tracer (Windows/Linux)
6. Implement communication between various network devices using CISCO packet tracer (Windows/Linux)
7. Network traffic monitoring using Wireshark/Ethereal (Windows/Linux)

List of Practical's: (Any Six)

1. Study and implement various networking commands on terminal.
2. Use Socket programming to create Client and Server to send Hello message.
3. Write a program for error detection and correction for 7/8 bits ASCII codes using HammingCodes or CRC. Demonstrate the packets captured traces using Wireshark Packet AnalyzerTool for peer-to-peermode. (50% students will perform Hamming Code and others willperform CRC)
4. Write a program to simulate Go back N and Selective Repeat Modes of Sliding WindowProtocol in peer-to-peer mode
5. Write a program to find class and type of a given IP address.
6. Write a program to demonstrate subnetting and find the subnet masks.
7. Write a program using TCP socket for wired network for following: a. Say Hello to Eachother (For all students) b. File transfer (For all students) c. Calculator (Arithmetic) (50%students) d. Calculator (Trigonometry) (50% students)
8. Write a program using UDP Sockets to enable file transfer (Script, Text, Audio and Videoone file each) between two machines.
9. Write a program to implement: a. Network Routing: Shortest path routing, AODV. b.Analysis of congestion control (TCP and UDP).
10. Write a program to analyse following packet formats captured through Wireshark for wirednetworks. 1.Ethernet 2. IP 3.TCP 4. UDP

List of Course Projects:
1. Write a program using TCP sockets for wired networks to implement a. Peer to Peer Chat b.Multi User Chat Demonstrate the packets captured traces using Wireshark Packet AnalyzerTool for peer-to-peer mode. 2. Implementation of shortest path protocol 3. Implementation of string encryption and decryption 4. Implementation of character stuffing and destuffing 5. Execution and analysis of Network commands 6. To find out details of network from IP addressing scheme using 'C' code 7. Implement real time Internet route optimization. 8. Implement Broadcast Server System. 9. Implement a real time voting System. 10. Real time packet capture and analysis for malwares in wireless networks.
List of Course Seminar Topics:
1. Asynchronous Transfer Mode 2. Need Of Multiplexing for Signal Modulation 3. TDM with PAM a case study 4. Noise signal 5. Basic Network Protocols 6. Manchester Vs Differential Manchester coding technique 7. Amplitude Shift Keying: Working and Applications 8. Nyquist Sampling Theorem 9. CDMA 10. Line coding Techniques with example
List of Course Group Discussion Topics:
1. TCP/IP Model 2. Mobile IP 3. Congestion Control and QoS 4. Wireless Technology for Short range and long range 5. Application Protocols and its security 6. IP Protocols 7. Data Communication Issues in IP Networks and Solutions to it 8. Congestion control in hybrid networks 9. Issues in Real time Audio and video transmission protocol. 10. IPV6
List of Home Assignments:
Design:

1. Enumerate the challenges in Line coding. Draw the line code for the sequence 010011110using Polar NRZ-L and NRZ-1 schemes.
2. 2.Design the procedure to configure TCP/IP network layer services.
3. Simulation of Routing Protocols using NS2
4. Simulation of FTP based Protocols using CISCO packet Tracer/ NS2
5. Simulation of Congestion Control Protocols Using NS2

Case Study:

1. Amplitude and Frequency Modulation Technique
2. Digital to Analog and Analog to Digital converters
3. Study of Various VPNs
4. IoT Solutions to Current Network Requirement
5. Unix Solutions for Broadcast System

Blog:

1. Communication Protocol
2. Emerging Trends in Computer Networks
3. Use of IOT in Networks
4. Cloud based Network Solutions for real world problems
5. Recent Trends in Computer Security

Surveys:

1. Survey of wireless Technologies
2. Survey of Congestion control methodologies
3. Survey of Bluetooth Technology
4. Survey of Virtual Private Networks
5. Survey of ADHOC Networks

Assessment Scheme:

PPT/GD
HA
ESE
Course Project
CVV

Text Books:

1. James F. Kurose, and Keith W. Ross," A Top-Down Approacht," 4th edition, Publisher: Addison-Wesley ISBN: 0-321-49770-8
2. Behrouz A. Forouzan ,"Data Communication and Networking", 4th edition, Tata McGraw Hill
3. Andrew S. Tanenbaum, "Computer Networks" ,5th Edition, Pearson Education

Reference Books:

1. Kurose, Ross,"Computer Networking a Top Down Approach Featuring the Internet‖", Pearson; 6th edition (March 5, 2012), ISBN-10: 0132856204
2. Holger Karl and Andreas Willig,"Protocols and Architectures for Wireless Sensor Network", Wiley, ISBN: 0-470-09510-5
3. C. Siva Ram Murthy and B. S. Manoj, "Ad Hoc Wireless Networks: Architectures and Protocols", Prentice Hall, 2004

Moocs Links and additional reading material:

1. www.nptelvideos.in
2. https://www.my-mooc.com/en/categorie/computer-networking

COURSE OUTCOMES

1. Select network architecture, topology and essential components to design computer networks.
2. Estimate reliability issues based on error control, flow control and pipelining by using bandwidth, latency, throughput and efficiency.
3. Design mechanisms to demonstrate server channel allocation in wired and wireless computer networks
4. Analyze data flow between peer to peer in an IP network using Application, Transport and Network Layer Protocols
5. Demonstrate Network Connections Strategies ,Protocols and Technologies
6. Develop Client-Server architectures and prototypes by the means of correct standards, protocols and technologies